

FISHING-INDUCED PREDATOR LOSS AND HABITAT DEGRADATION RESHAPE REEF FISH BEHAVIOUR

Authors: Shawn Dsouza, Akshita Joshi, Samar Ahmad, Bharat Ahuja, Titus Immanuel, Maria Thaker, Kartik Shanker.
Centre for Ecological Sciences, Indian Institute of Science, Bengaluru, India.



INTRODUCTION

In the Anthropocene, humans disproportionately target top predators, leading to "**trophic downgrading**."

In marine ecosystems, the loss of predators doesn't just change who is on the reef; it changes how the remaining fish behave.

This study explores the altered "**Seascape of Fear**" in coral reefs around the Andaman Islands.

Questions:

How does predator removal by fisheries alter prey behaviour?

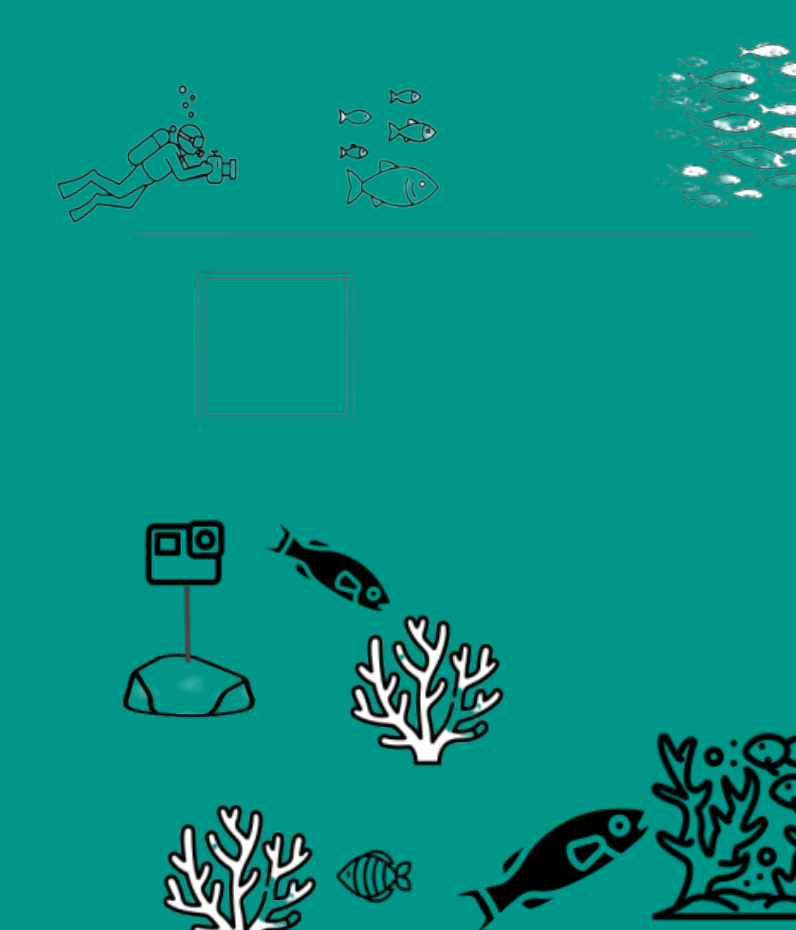
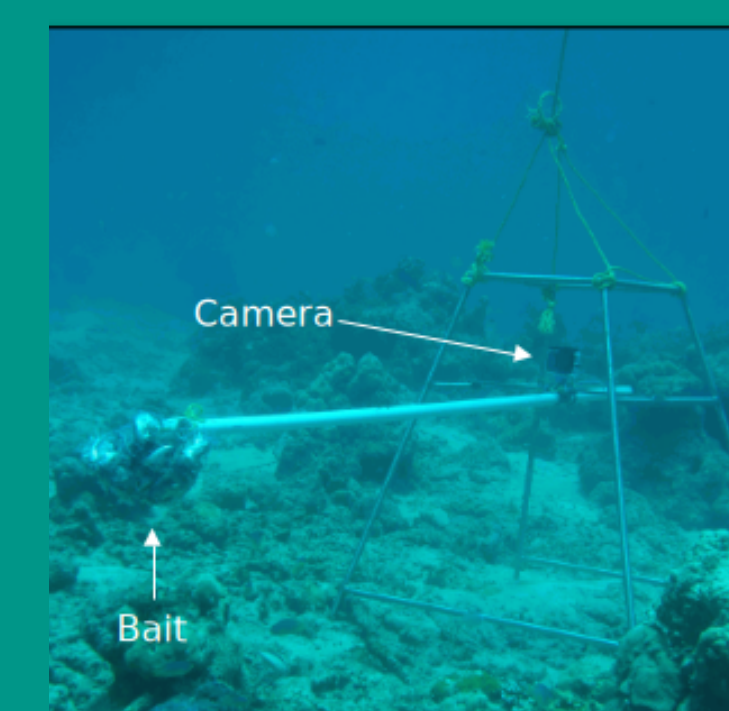
How do fishery-driven habitat changes (structure and resources) shape prey behaviour?

Do prey populations “forget” predators after prolonged human-driven predator loss?

Hypothesis:

Anti-predator responses will be suppressed outside marine protected areas (MPA) relative to inside MPAs.

METHODS



Study sites: Inside and outside two MPAs in **South Andaman** (MGMNP, RJMNP).

Predators: BRUVS to estimate piscivore abundance and size.

Habitat: Photo-quadrats to quantify benthic cover, structural complexity, and resources.

Behaviour: RUV to measure **foraging**, **vigilance**, and **movement**.

Experiments: **3D-printed predator models** (leopard grouper = ambush; great barracuda = pursuit) to test risk recognition.

Analysis: Bayesian causal inference and counterfactual simulations.

RESULTS AND DISCUSSIONS

Predator loss alters prey behaviour and time budgets

Herbivorous and invertivorous fishes spent 1.5–2× more time vigilant inside MPAs than outside.

These fishes showed ~0.5× lower movement rates inside MPAs.

Foraging rates were higher inside MPAs: 1.25× (herbivores) and 2× (invertivores).



Ambush Predator : Leopard Grouper
(*Mycteroperca rosacea*)

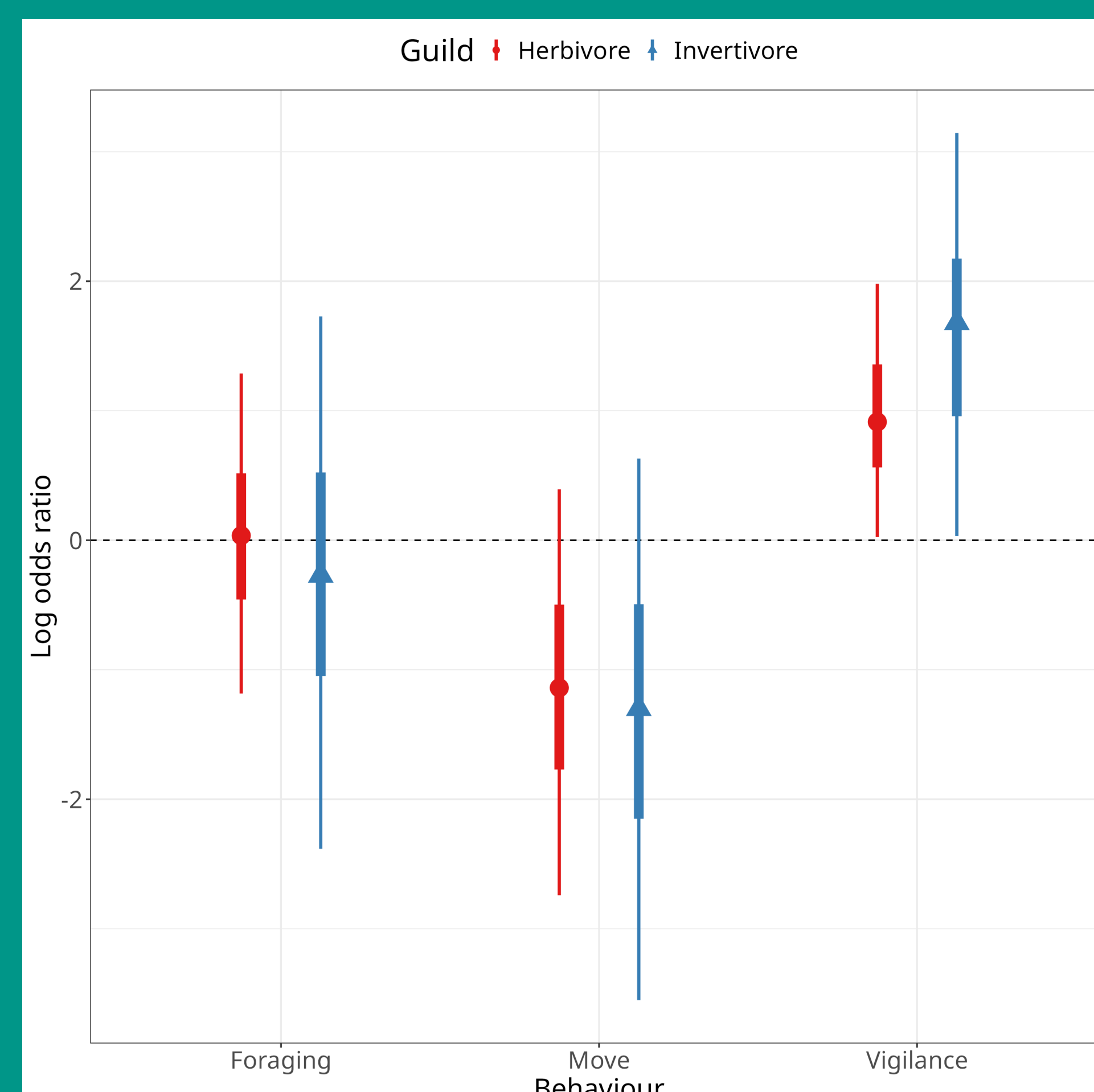


Coursing Predator : Great Barracuda
(*Sphyraena barracuda*)

MPAs maintain higher predator biomass and healthier habitats

Predator abundance was ~1.5× higher inside MPAs than outside.

Coral cover was 16% higher within MPAs, indicating greater habitat complexity.



Prey responses are context dependent

Fish preferred more structurally complex habitats, but showed higher vigilance there.

Grouped individuals were less vigilant and foraged longer than solitary individuals, both inside and outside MPAs.



“Out of sight, out of mind”: predator recognition is weakened outside MPAs

Fish inside MPAs increased vigilance to both predator models; fish outside showed little or no response.

Herbivores responded more to groupers (ambush); invertivores to barracudas (pursuit).

